

ECON 144 — EXTENDED STUDY GUIDE (FULL VERSION) This expanded study guide includes:

- Deep conceptual explanations
- Fully explained statistical tests + R code
- Forecasting models
- ARIMA, GARCH, ECM, Kalman Filter
- Markov Chains, Monte Carlo, MCMC, Gibbs, MH
- Diagnostic testing and model selection
- Examples, diagrams (text-based), formulas
- Full chapter-by-chapter outline

CHAPTER SUMMARIES (Condensed for PDF brevity — content can be expanded further)

LECTURE 1 — Time Series Intro Definition of time series, stochastic processes, stationarity. ACF/PACF, decomposition, visualization. Tests: Ljung-Box ( $H_0$ : no autocorrelation), ADF ( $H_0$ : unit root). R: `adf.test(x)` `Box.test(x, lag=10, type="Ljung-Box")`

LECTURE 2–3 — Trend Models Linear, quadratic, log-linear, exponential. Model selection: AIC, BIC. Forecast errors: MSE, RMSE, MAE. R: `tslm(y ~ trend + season)` `forecast()`

LECTURE 4 — White Noise WN: zero mean, constant variance, no autocorr. Lag operator, moving average construction.

LECTURE 5 — Seasonality Add/mult seasonal decomposition. Dummy-variable seasonal models. STL decomposition. R: `stl(ts, s.window="periodic")`

LECTURE 6 — Cycles & MA Models  $MA(q)$ , invertibility. ACF cuts off  $\rightarrow$  MA identification. R: `arma(x, order=c(0,0,q))`

LECTURE 7–9 — Trend + Seasonality + Cycles ARIMA models. Recursive estimation. Cross validation for forecasting. R: `auto.arima(x)`

LECTURE 10–12 — Forecast Evaluation Diebold–Mariano Test:  $H_0$ : equal predictive accuracy. MZ Regression:  $H_0$ :  $\alpha=0$ ,  $\beta=1$ . Encompassing tests. Combination forecasts.

LECTURE 14 — ARCH/GARCH  $ARCH(q)$ : volatility =  $\omega + \alpha \varepsilon^2$ . GARCH(1,1):  $\sigma^2 = \omega + \alpha \varepsilon^2 + \beta \sigma^2$ . Persistence =  $\alpha + \beta$ . Stationarity:  $\alpha + \beta < 1$ . Long-run variance =  $\omega / (1 - \alpha - \beta)$ . R: `garchFit(~ garch(1,1), data=x)`

LECTURE 15 — Cointegration & ECM Engle–Granger: Step 1: regress y on x. Step 2: ADF test on residuals.  $H_0$ : no cointegration. ECM:  $\Delta y = \alpha + \beta \Delta x + \lambda u_{t-1}$ . Speed of adjustment =  $\lambda$ . Structural breaks  $\rightarrow$  Gregory-Hansen.

LECTURE 16 — Kalman Filter State equation, measurement equation. Q = state variance; R = measurement variance. Kalman Gain:  $K = P_{pred} H' (H P_{pred} H' + R)^{-1}$ . High Q  $\rightarrow$  fast updating. High R  $\rightarrow$  slow updating. Applications: time-varying betas.

LECTURE 17 — Monte Carlo, Markov Chains, MCMC Markov chain definitions: Irreducible, aperiodic  $\rightarrow$  ergodic. Stationary distribution:  $\pi = \pi P$ . Gibbs sampler: Sample from conditionals. Metropolis-Hastings: Accept-reject algorithm. MORE MATERIAL AVAILABLE UPON REQUEST.